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IEEE Open Journal of Antennas and Propagation

Received XX Month XXXX; revised X Month XXXX; accepted XX Month XXXX. Date of publication XX Month XXXX; date of current version XX Month XXXX.

Digital Object Identifier

60 GHz Programmable Dynamic Metasurface Antenna (DMA) for Next-Generation Communication, Sensing, and Imaging Applications: From Concept to Prototype

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This work was supported by Innovative and Collaborative Research Grant under Pakistan-UK Education Gateway (ICRG-2020) Project No. 310366-Deforestation in Pakistan: Combating through Wireless Sensor Networks (DePWiSeN), and by EPSRC projects, CHEDDAR EP/X040518/1 and CHEDDAR Uplift EP/Y037421/1

ABSTRACT In this paper, for the first time we present the complete design of a dynamic metasurface antenna (DMA) array at the 60 GHz millimeter-wave (mmWave) industrial, scientific, and medical (ISM) band. First, a novel complementary electric inductive-capacitive (CELC) metamaterial element (unlike conventional rectangular CELC) is designed to resonate around 60.5 GHz. The proposed CELC meta-element in its resonance state manifests dispersive characteristics and exhibits significant left-handed metamaterial properties such as negative group refractive index, negative effective permittivity, and negative group velocity, which are thoroughly elucidated. A low-loss V-band planar substrate-integrated waveguide (SIW) structure is designed at dominant TE_{10} to excite the CELC meta-element by an in-plane magnetic field. Two PIN diodes are loaded in the small capacitive gap between the CELC meta-element and the SIW structure. The switching state of the PIN diodes readily renders the meta-element either radiating or non-radiating. The difference between radiating and non-radiating states is more than 11 dB. Consequently, a fully addressable digital tunable DMA element is formed. Then, a one-dimensional DMA is designed by embedding 16 such meta-elements into the upper conducting wall of the edge-fed SIW structure for electronic steering with high gain, high radiation efficiency, and low side lobe levels. The radiation state of each CELC meta-element is dynamically controlled through a high-speed field programmable gate array (FPGA). The DC biasing network for PIN diodes at such high frequency is meticulously designed and integrated using 4-layer standard printed circuit board (PCB) technology. The parallelized biasing network of PIN diodes through a high-speed FPGA enables agile dynamic control over the radiation pattern of the entire digitally coded metasurface aperture. Versatile beam synthesis (such as narrow beams, wide beams, and multiple beams) is achieved based on different digital coding combinations. The prototypes of the single DMA element and 16-element 1-dimensional DMA array are fabricated and verified through practical measurements. Simulated and measured results show good agreement. The beam-switching agility is quantified and observed to be within 5 ns, indicating significant promise for mmWave applications with ultra-low latency. The proposed DMA is a potential enabler to unfold a diverse range of next-generation mmWave wireless applications such as agile electronic beam-steering, adaptive beamforming and beam-shaping, holographic computational imaging, mmWave industrial wireless communication, cognitive radars, as well as integrated sensing and communication (ISAC).

INDEX TERMS 60 GHz ISM band, adaptive beamforming, dynamic metasurface antenna, electronic beam-steering, holographic antenna, smart antenna, SIW.

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I. INTRODUCTION

The remarkable properties of metamaterials to manipulate electromagnetic (EM) waves have yielded a variety of exotic phenomena and gained significant interest in a variety of domains. Their effective parameters can be tailored when used within planar structures, also known as metasurfaces, to accomplish a specific transformation of transmitted, received, or incident EM waves. The next-generation (beyond 5G and 6G) wireless technologies are envisioned to adapt to dynamic conditions through intelligent and software-reconfigurable paradigms to maintain the quality of service and seamless connectivity. Therefore, much of the attention has been paid to the dynamically tunable metasurfaces whose operational status can be electronically reconfigured, such as reconfigurable intelligent surfaces (RIS) and coded reflect/transmit metasurface arrays [1]–[6]. Mainly, the dynamic coded surfaces work on the principle of reflections and are required to be fed through an external source antenna which renders the whole network quite bulky [7]–[9]. A comprehensive review of reflective digitally coded metasurfaces is presented in [10]. Nevertheless, instead of reflections, the radiative type metasurfaces have recently gained attention as they possess the benefits of in-plane circuit feed, low profile, dynamic tunability, and seamless integration with the radio-frequency (RF) frontends.

Dynamic metasurface antenna (DMA) is an emerging concept under the paradigm of "radiative metasurfaces" which offers controllable radiation pattern diversity and adaptability from a simplified and compact hardware platform [11]. Some other nomenclatures are also used in literature, such as Reconfigurable Holographic Surfaces [12]–[15]. Predominantly, these metasurface antennas make use of polarizable dipole metamaterial elements of the sub-wavelength size that can dynamically reconfigure their radiation pattern using a single channel, thereby providing efficient control over steerable beams [16]–[18]. A conceptual representation of the waveguide-fed one-dimensional (1-D) DMA is shown in Fig. 1. The metamaterial elements couple with a portion of the guided mode that propagates inside the waveguide and radiate it into the free space. Positive-intrinsic-negative (PIN) diodes integrated with each meta-atom, independently control the coupling and non-coupling states of each element through an external programmable circuit, such as Field Programmable Gate Array (FPGA). Not only does this transformational feature improve signal quality and beam pattern, but it also allows adaptive beamforming, and pattern diversity, making these antennas desirable for next-generation mmWave communication, sensing, and imaging (CSI) applications [5], [11], [13], [19], [20].

A DMA is based on metamaterial elements consuming substantially less power and offers low-cost than conventional phased array antenna systems because it eliminates the need for complex and lossy corporate feed systems, as well as active phase shifters and amplifiers. Furthermore, because a metamaterial element's resonance can be dynamically tuned using a variety of modalities, including semiconductor components such as PIN diodes, varactors as well as liquid

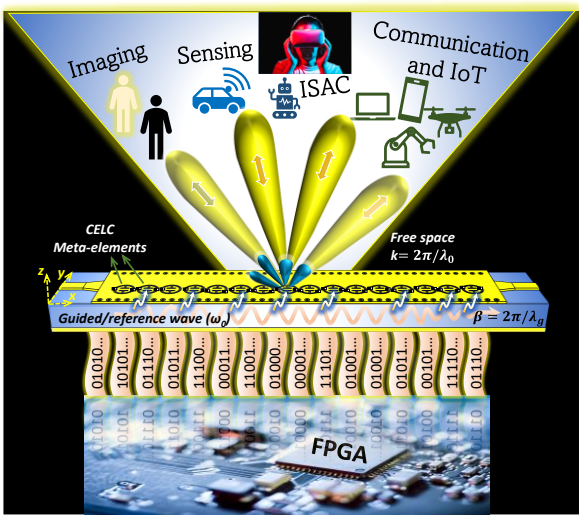


Fig. 1. A conceptual depiction of a DMA integrated with a software-programmable FPGA. The excitation states "0" and "1" of the meta-elements are independently controlled according to the applied 1-bit digital coding sequence through embedded PIN diodes, resulting in the desired real-time agile electronic beam-steering.

crystals, a DMA can be tuned with extremely low power requirements and without the need for complex active circuitry [18]. The remarkable features such as smaller size, lower cost, easy integration, and feed mechanism through planar waveguide apertures, reduced hardware complexity, and lower power consumption make DMA an interesting and viable choice for the next-generation of smart wireless networks [11], [12], [14], [21], [22].

The dynamic metasurface apertures have been employed for imaging and sensing applications [19], [20], [23]–[26]. Moreover, owing to their promising potential, recently some of their other EM aspects have been demonstrated such as channel estimation [27], communication model [28], and analysis using discrete dipole framework [29]. Although much research interest has been laid in recent years towards sensing and imaging applications of DMAs, however, a promising higher mmWave hardware-level design of DMA to serve as a high-performance beam-steering antenna array for the wireless communication domain is highly desirable and is still in its infancy. Quite a few DMA prototypes have been reported for CSI applications in the microwave band up to 30 GHz, such as around X-band (8 to 12 GHz) [30]–[32], 23.5 GHz [33], and 27 GHz [34], using conventional rectangular CELC resonators. Nevertheless, owing to very small dimensions at higher mmWave bands, the DMA design along with integrated RF feed, biasing circuitry, and controllable hardware becomes immensely challenging. More specifically, the design and experimental verification of DMAs at higher mmWave bands (around 60 GHz) and sub-THz bands is a critical next step in unlocking their potential for 6G communication and is still in the early phases of research, as explicitly envisioned in [11].

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In this paper, for the first time, we present a meticulously designed programmable DMA at 60 GHz mmWave band, controlled through a high-speed FPGA. We thoroughly elucidate the complete design philosophy, construction, and working principle of the proposed mmWave DMA element as well as 1-D array from concept to prototype. The main contributions of this article are as follows:

- First, we proposed the design of a novel CELC meta-element geometry (as opposed to the commonly used rectangular CELC element) at 60 GHz mmWave band. The complete design philosophy of the proposed CELC meta-element is elucidated. The dispersive characteristics and left-handed metamaterial properties of the proposed CELC meta-element are thoroughly investigated and presented through numerical simulations and verified through experiments.
- We designed a fully addressable and dynamically tunable DMA element based on the proposed CELC meta element. Realistic simulations were carried out including the complete biasing network and PIN diodes which reveal the minimal effect on RF performance. A low-loss substrate integrated waveguide (SIW) structure is designed, and the CELC element is edge-fed using a microstrip line for ease of measurement and ease of integration with RF circuitry.
- We meticulously designed a 4-layer PCB of DMA prototype comprising 16 CELC meta-elements spaced at Nyquist limit ensuring high gain, low side lobes, high radiation efficiency, and compact profile with more than 2.16 GHz of -10 dB impedance bandwidth. The array is integrated with high-speed FPGA for dynamic control of the radiation state of each meta-element in real-time. All the PIN diodes (and thus all meta-elements) are parallelized, therefore agile beam-switching is achieved by having a latency level within 5 ns, paving the way towards extremely low latency mmWave communication.
- The DMA is fully controllable through software programming and can produce fixed-frequency electronic beam-steering. Our proposed DMA is capable of synthesizing various steerable radiation patterns including narrow beams, wide beams as well as multi-beams based on different digital coding sequences using a single antenna aperture. This immensely reduces the hardware and signal processing complexity by avoiding the use of multiple antennas for imaging and sensing applications.
- With a notion of adaptability to dynamic reconfigurability of EM environment for beyond 5G/6G technologies, we designed the DMA prototype in an extremely flexible and programmable manner to work with various beamforming algorithms in the communication domain, as well as signal processing algorithms in the sensing and imaging domain. The versatility of our 60 GHz DMA prototype provides a readily accessible and off-the-shelf solution to explore and investigate a diverse range of next-generation mmWave applications.

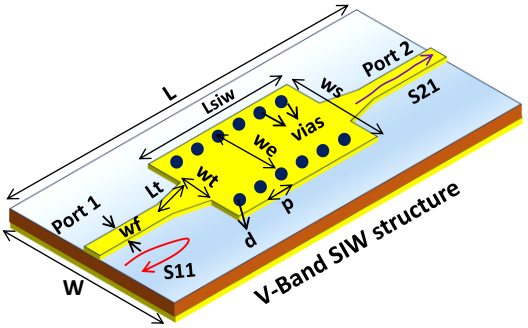


Fig. 2. Schematic design of V-band SIW structure. (dimensions in mm). L=20.32, W=15, wf=0.62, Ls = 4.16, ws = 4.16, we = 3.05, wt = 1.25, Lt = 2.16, d = 0.2, p = 0.4, Lsiw = 4 for DMA element and Lsiw = 27.36 for DMA.

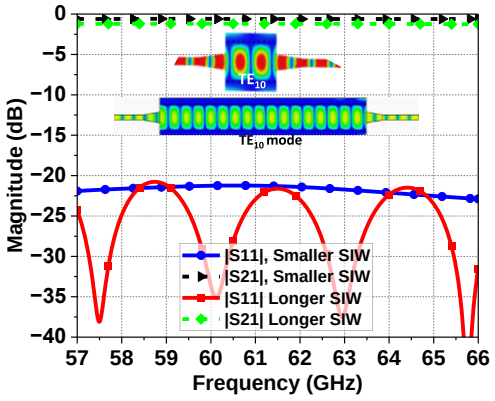


Fig. 3. Reflection and transmission magnitude of designed SIW structure.

II. DESIGN METHODOLOGY OF THE PROPOSED DMA

In general, a DMA primarily consists of numerous resonating metamaterial elements placed periodically in a planar waveguide structure. We divide the design steps of a DMA into two main steps. First, we present the design of a planar SIW structure which will be used to excite the metamaterial element. This is because the CELC resonator is a magnetic dipole and thus magnetic field (H-wave) is required to excite it, whereas the planar SIW structure readily provides TE_{mn} modes which are H-waves. The second step is the design of the CELC resonator at the desired frequency band and integrating it into the waveguide aperture. The pattern reconfigurability can then be achieved by forming a multitude of meta-elements and incorporating passive electronic components, such as PIN diodes with each CELC resonator along with an appropriate biasing network to achieve beam-steering.

A. DESIGN OF 60 GHz (V-BAND) SIW STRUCTURE

The SIW structure was designed using Rogers 3003 substrate, with a dielectric constant of 3, thickness of 0.25 mm, and copper cladding of 17.5 microns, as shown in Fig. 2. An electroplated wall of vias is designed carefully to achieve high

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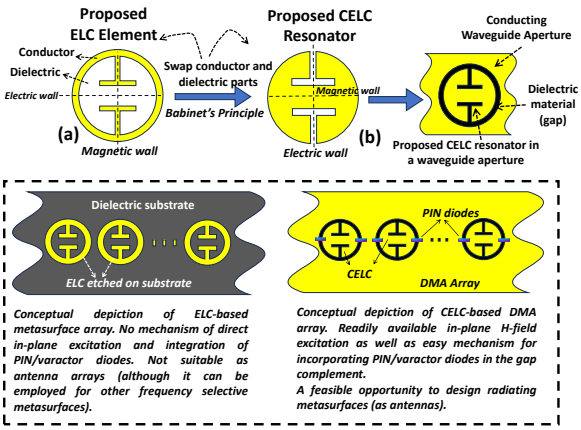


Fig. 4. Design philosophy of the proposed CELC metamaterial unit cell resonator.

transmission (S21) and low reflection coefficient (S11) around 60 GHz band. The diameter of vias (d) is set as 0.2 mm (0.04 λ_0) and the distance (pitch) between any two consecutive vias (p) is set as 0.4 mm (0.08 λ_0) to minimize the leakage losses (here λ_0 is the free space wavelength at 60 GHz) [35]. We designed the SIW structure to operate at the first dominant TE_{10} mode. The cutoff frequency of the dominant mode and effective width of SIW is calculated as:

$$f_c(TE_{10}) = \frac{c}{2\sqrt{\epsilon_r} w_{eq}} \quad (1)$$

$$w_{eq} = w_{siw} - \frac{d^2}{0.95p} \quad (2)$$

Here c is the speed of light, w_{siw} is the distance between two rows of vias, and w_{eq} is the effective width of SIW which is a function of the pitch length and the diameter of vias. These dimensions were then further optimized in an electromagnetic solver to achieve high S21 and low S11 parameters to make sure that the SIW structure itself should not radiate. A 50 Ω microstrip feed line having a width of 0.62 mm is designed and a tapered transition from microstrip to SIW is optimized with the proposed width of 1.25 mm to ensure better reflection and transmission performance. The edge-fed SIW makes the design extremely easy for measurements as well as to integrate it with other RF radio circuitry, such as with mmWave power sensors for RF sniffing.

The magnitudes of the reflection coefficient (|S11|) and transmission coefficient (|S21|) of the SIW structure are shown in Fig. 3. Once carefully designed, the length of SIW does not significantly affect the reflection coefficient while insertion loss is merely added for longer structure (i.e., |S21| can be reduced due to resistive and dielectric losses of the longer substrate), thus it can be scaled down in length to excite a single CELC resonator, as required. For single CELC resonator excitation, SIW length (L_{siw}) is 4 mm, whereas to accommodate 16 CELC resonators, the overall L_{siw} is 27.36 mm. The |S21| (insertion loss) is better than –1.19 dB for longer SIW and -0.6 dB for shorter SIW, whereas |S11| is below -21 dB in both cases in the desired band of interest, covering the entire 57-66 GHz ISM band. The phase distribution in the structure is uniform and the E-field distribution of the SIW structure illustrates TE_{10} mode confined with the SIW structure, as shown in the inset of Fig. 3.

B. DESIGN OF THE PROPOSED 60 GHz CELC METAMATERIAL UNIT CELL

The design philosophy of a DMA element is demonstrated in Fig. 4. Consider first the proposed design of an ELC element in Fig. 4(a). The structure consists of a conducting ring with two mutually perpendicular strips at two positions to create a small capacitive gap between them which responds to an electric field (E-field). The ELC elements radiate poorly when embedded in a conducting plane and therefore are not a feasible choice for radiating metasurfaces (i.e., antenna arrays). Having said that, the proposed ELC resonator can be employed to design frequency-selective metasurfaces for other applications, which is not the scope of this work. On the other hand, their complement metamaterial elements (i.e., CELC) with an effective magnetic response provide better radiation characteristics [18]. According to Babinet's principle [36], the dual/complement of an ELC element (i.e., CELC) is a pure magnetic resonant structure that couples strongly to the magnetic field (H-field) [17], [37]. The proposed CELC resonator is designed by interchanging the conductor and dielectric parts of the ELC, thereby swapping electric and magnetic walls, as shown in Fig. 4(b).

The design and geometry of the CELC element are a crucial part of the DMA design and primarily dictate the spectral region of operation. While some intuition may be used during the design stage of a DMA, the characteristics of the element are not individually modified through a single dimension. Instead, an extensive optimization approach based on full-wave numerical simulations is required to achieve the desired radiation features. The design involves careful optimization of the geometry and sub-wavelength size to achieve its resonance response at a desired frequency. The dimensions of the inset cuts are optimized to resonate around 60.5 GHz and play a crucial role in the efficient radiation performance of the CELC element. The diameter of the CELC element is 0.9 mm (λ_0/5.55). A circular slot with a diameter of 1.1 mm is etched in the SIW aperture to place the designed CELC resonator. The position of CELC is horizontally equidistant from both ports for symmetry, while the vertical offset was optimized at 0.5 mm from the center for better DMA performance. The radial gap between the SIW aperture and CELC is 0.1 mm for mode coupling and loading the PIN diodes effectively. The DMA is fed from port 1, while port 2 is used to dampen the residual energy in the structure and avoid standing waves. The proposed schematic of the DMA element is shown in Fig. 5(a).

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C. TUNABILITY OF CELC META-ELEMENT AND BIASING SCHEME

The tunability of the CELC resonator is achieved by controlling the switching state of the loaded PIN diodes (MADP-000907-14020W) on the capacitive gap between the CELC and SIW aperture. In simulations, the PIN diode was modelled as an equivalent lumped element circuit as shown in Fig. 5(b). For the forward biased state (diode on), a series combination of L = 30 pH and Rs = 5.2 Ω was used, whereas for the reversed biased state (diode off), a combination of L = 30 pH with Cp = 25 fF and Rp = 10 kΩ was used.

The cross-sectional view of the printed circuit board (PCB) layer stack of the DMA consists of 4 layers, as presented in Fig. 6(a). The top layer (L1) comprises CELC meta-element and SIW conductor, layer 2 (L2) includes RF ground, layer 3 (L3) is exclusively prepared for +DC power and radial stubs, while layer 4 (L4) consists of biasing lines, DC ground area, components (such as voltage regulator integrated circuits (ICs) and coupling capacitors to reduce switching noise, and light-emitting diodes (LEDs) for debugging and beamforming visualization. Further detail about layer stack-up is provided in Section V-A.

The PIN diodes are biased with a positive DC voltage at the center of the CELC resonator through a metalized via of 0.2 mm diameter, extending from the center of the CELC (L1) to the DC control line at the bottom (L4). The choice of the center point of CELC for biasing via is because it indicated near zero E-field distribution in the numerical simulations, therefore its impact on the radiation pattern and the guided wave is minimal while biasing. The DC and RF share a common ground layer and the DC negative terminal can be connected to the external metallic body of the feed connector during practical antenna measurements. To separate the DC line from RF, an open-ended radial stub of quarter-wavelength (at 60 GHz) radius was designed with biasing via at L3. Therefore, by generating a virtual short circuit for RF current, this sectoral radial stub effectively blocks RF from reaching the DC source. These methods minimize the additional loss produced by the biassing network, which often occurs in reconfigurable reflect arrays. The realistic schematic model of a single DMA element including PIN diodes and DC biasing network (i.e., biasing via, radial stub and DC control line) is illustrated in Fig. 6(b), whose realistic simulated results are presented in Section IV.

D. RADIATION MECHANISM AND THEORY OF OPERATION

The biasing state of the PIN diode decides the radiating (coupling) or non-radiating (non-coupling) state of the CELC resonator. When the diode is forward-biased, the CELC resonator is effectively shorted with SIW aperture and thus becomes a part of the SIW structure. As a result, its resonance and radiation characteristics are lost. On the other hand, when the PIN diode is reverse-biased, the CELC resonator is effectively isolated from the SIW structure and becomes analogous to an open circuit with respect to the SIW aperture. Consequently, the CELC resonator couples the

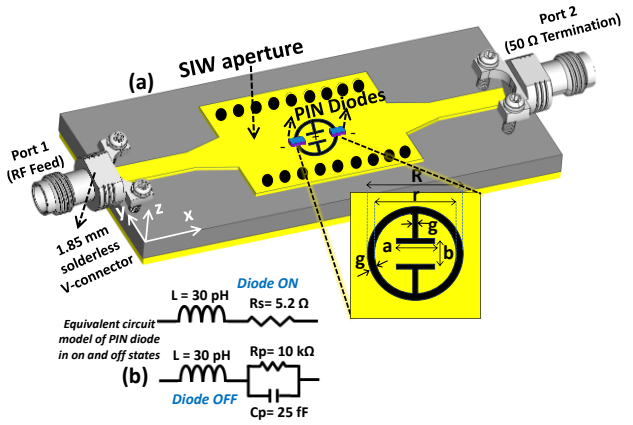


Fig. 5. (a) Schematic design of the proposed SIW-fed DMA element. Only the antenna part is depicted in this figure. (R = 1.1, r = 0.9, g = 0.1, a = 0.3, b = 0.2). Dimensions are in mm. (b) Equivalent circuit model of PIN diode in on and off states.

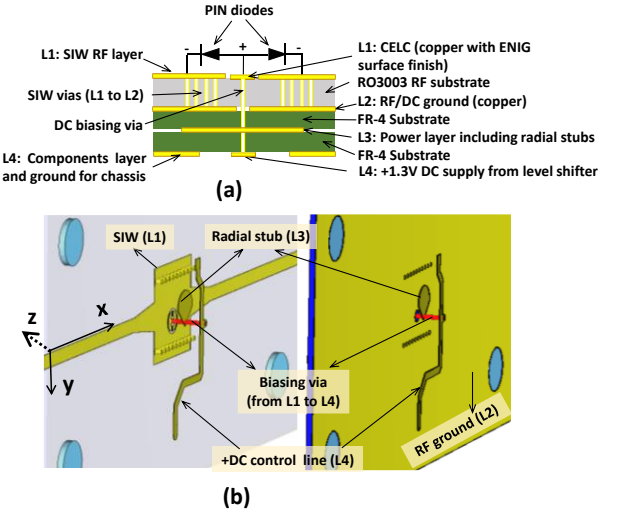


Fig. 6. (a) Cross-sectional 4-layer PCB stack up view (b) A realistic schematic model of the single DMA element including biasing network for accurate simulations. (left) without RF ground shown (i.e., L2). (right) with RF ground (L2) shown.

portion of the waveguide mode to the radiation mode and exhibits strong radiation characteristics. In this way, a 1-bit "digital element" with either radiating ("1") or non-radiating ("0") state forms the basis of a DMA.

Note that in this mode of operation, the diode-off state leads to the CELC element-on state, and the diode-on leads to the CELC element-off state. Therefore, the overall DMA design in the radiation state consumes much less power as compared to the conventional power-hungry phased array systems. Eventually, the binary digital coding combinations ("0s" and "1s") when applied to a number of CELC resonators in an array topology form the basis of a digital programmable metasurface antenna array for fixed frequency electronic beam-steering, controlled externally through a high-speed FPGA.

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III. DISPERSIVE CHARACTERISTICS AND LEFT-HANDED METAMATERIAL PROPERTIES OF THE PROPOSED CELC META-ELEMENT

There has been a lot of interest in artificially engineered materials with effective negative permittivity (ENG), effective negative refractive index (NRI), and negative group velocity (NGV) after the concept was theorized by Veselago [38]. These media, also known as left-handed metamaterials, exhibit unique electromagnetic features that are not observed in naturally occurring materials, such as the reversal of Snell's Law, the Doppler effect, and the Vavilov-Cerenkov effect (radiation produced by a fast-moving particle as it travels through a medium) and unusual focusing properties [39], [40]. What distinguishes the resulting media is that the effective permittivity, effective refractive index, and group velocity can attain negative values around the resonant frequency band with high dispersion characteristics (known as anomalous dispersion) which are not observed in conventional materials.

The proposed CELC element in the radiation state exhibits highly dispersive properties of the DMA structure which are elucidated here. These dispersive characteristics depend on a strong absorption spectrum around the resonance and can be extracted from the S-parameters of the structure. The transmission phase around the resonance further elucidates the resonance phenomenon. The double phase slope reversal within the absorption resonance band exhibits anomalous dispersion characteristics of the DMA element during its radiation state, as shown in Fig. 7.

The anomalous dispersive phase is attributed to negative group delay (τ_g), negative group refractive index (n_g), negative group velocity (v_g), and negative slope of the refractive index (n) within the resonance band. The concept of group delay is closely related to group velocity, and provides a way to describe the velocity of propagation of electromagnetic wave packets [41]. For a well-behaved wave packet, it is the time delay that the signal envelope experiences as it travels through a medium of specified length. It is related to the negative of the angular frequency (ω) derivative of the transmission phase (ϕ_{S21}) as:

$$\tau_g = -\frac{d\phi_{s21}}{d\omega} \tag{3}$$

It can be observed from Fig. 7 that the slope of the S21 phase is positive around the resonance, which according to (3) leads to a negative group delay. The negative of the radiating CELC element around its resonant frequency is shown in Fig. 8 which is of the order -0.53 ns. The group velocity and group refractive index are related as:

$$v_g = \frac{c}{n(\omega) + \omega(\frac{dn}{d\omega})} \tag{4}$$

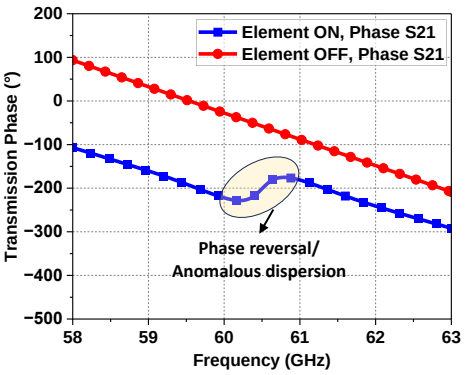


Fig. 7. Transmission phase of the proposed DMA element in radiating (element-on/ "1") and non-radiating (element-off/ "0") states.

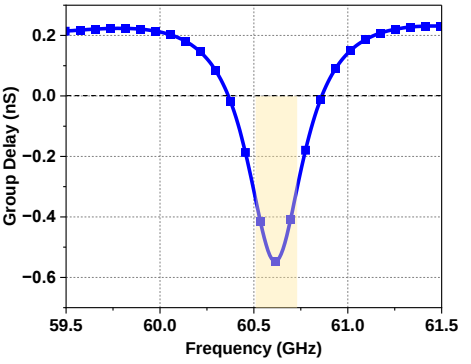


Fig. 8. Group delay of the radiating CELC meta-element.

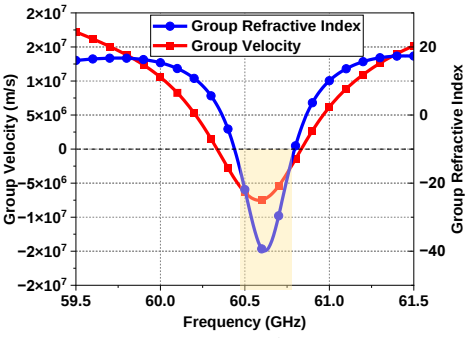


Fig. 9. Group velocity and group index of the radiating CELC meta-element.

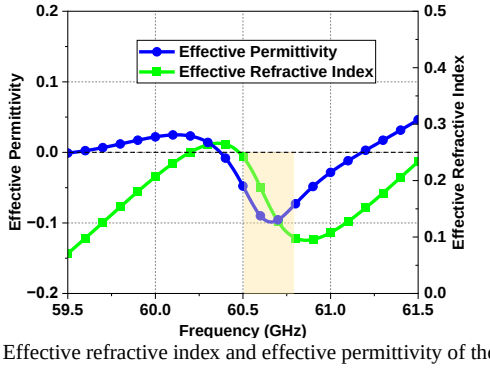


Fig. 10. Effective refractive index and effective permittivity of the radiating CELC.

In (4), c is the speed of light in vacuum and the whole term in the denominator refers to the group refractive index.

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According to (4), the decreasing value of the refractive index with frequency (i.e., negative slope) leads to the negative group velocity and hence the negative group index (n_g). It can be noted from Fig. 9 that around the resonance band of the radiating CELC element, the group velocity and group index are negative. This is in agreement with (4) by noting that the slope of the effective refractive index around the resonance band is negative, as shown in Fig. 10. Moreover, as demonstrated in Fig. 10, the effective permittivity is also negative around 60.5 GHz. These characteristics exhibited by the proposed CELC element satisfy the properties of a left-handed metamaterial medium [39].

IV. SIMULATED RESULTS AND ANALYSIS OF SINGLE DMA ELEMENT

A. S-PARAMETER ANALYSIS IN RADIATING AND NON-RADIATING STATES

The magnitude of S11 and S21 of the proposed DMA element in on and off states is shown in Fig. 11. When the PIN diode is off, the CELC couples to the waveguide mode and absorbs energy which is then radiated. The element-on state of the CELC shows strong Lorentzian-like resonance at 60.5 GHz characterized by |S21| dip of -15 dB. Note that around the resonance band where S21 drops, the CELC element being an open circuit resonator offers high impedance to the input side and therefore S11 might degrade. However, being an antenna property, it is always preferable to maintain |S11| to be less than -10 dB to maximize the accepted power and minimize standing waves. In the proposed design, |S11| of the DMA element in the radiating state is less than -10 dB around the resonance band. When the PIN diode is on, CELC behaves as a short circuit and virtually becomes part of the SIW structure. Consequently, its resonance effect is lost, and its impedance is matched to that of the SIW structure therefore maximum energy flows towards port 2, thereby presenting |S21| > -1.5 dB and |S11|< -15 dB. It is important to note that a single CELC element manifests a high Q-factor and narrow bandwidth of operation, however, with array configuration, the overall operating bandwidth is enhanced by resonance combination of multiple CELC meta-elements.

It is instructive to mention here that the proposed DMA element prototype without loading the PIN diode acts as an ideal DMA element with an open circuit CELC resonator that will always be a radiating case around 60.68 GHz, due to coupling with waveguide mode through the open circuit capacitive gap. Here we present the effect of loading PIN diodes with the CELC element as well as the effect of biasing network to assert the most realistic simulation results. Thus, we present the simulation results for four different design cases to test a single DMA element: 1) open circuit DMA element, without biasing and without diodes, which represents the most ideal case to analyze the resonance behavior of CELC while reducing the simulation time. 2) Design with loaded PIN diodes but without a biasing network. 3) Inclusion of biasing network but without PIN diodes. 4) DMA with complete biasing network as well as loaded reverse-biased PIN diodes (i.e., radiating CELC).

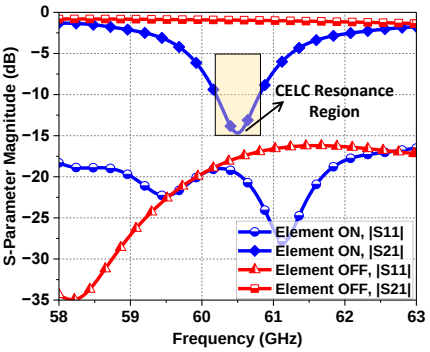


Fig. 11. S11 and S21 of the proposed DMA element in radiating (element ON) and non-radiating (element OFF) states.

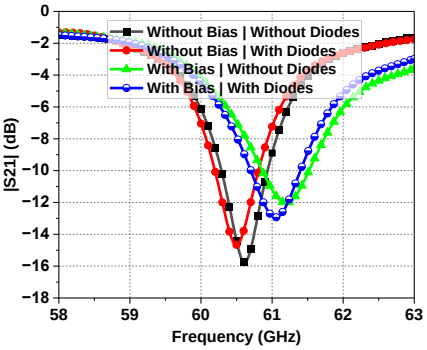


Fig. 12. |S11| and |S21| of the proposed single DMA element in radiating and non-radiating states.

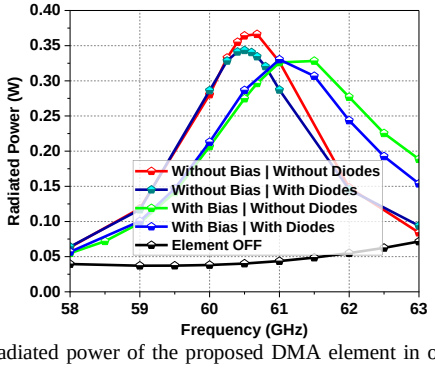


Fig. 13. Radiated power of the proposed DMA element in on and off states.

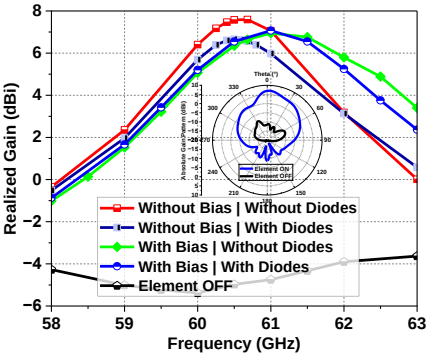


Fig. 14. Realized gain of the proposed DMA element in on and off states.

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The impact of the above-mentioned design cases on the resonant frequency (|S21| shift) is presented in Fig. 12.

It can be noted that the resonance frequency shifts from 60.6 GHz to 61 GHz when the biasing network is included in the model. This blue shift is due to the addition of parallel inductance of the biasing via with inherent inductance of CELC meta-element (note that the net inductance decreases in a parallel combination, $\omega_0 = \frac{1}{\sqrt{LC}}$). Similarly, the loaded diodes cause a slight redshift in the resonance frequency due to the addition of small parasitic capacitance of PIN diodes in the off state (note that the net capacitance increases in a parallel combination). Typically, the resonance band radiated power, and realized gain showed minimal impact from the inclusion of a biasing network and PIN diodes. This ensures the effective and careful design considerations employed in our proposed 60 GHz DMA element, which would serve as a building block of a large DMA.

B. RADIATED POWER, REALIZED GAIN AND RADIATION EFFICIENCY OF DMA ELEMENT

The power profile of the DMA element in on and off states is shown in Fig. 13. For an input power of 0.5 W and accepted power of 0.47 W, the peak radiated power of DMA in the radiating state is 0.36 W at 60.68 GHz for open circuit CELC (i.e., without embedding PIN diodes and bias network). With the inclusion of a bias network and PIN diodes, the peak radiated power of 0.33 W is observed at 61 GHz with the accepted power level of 0.45. Thus, the radiation efficiency of the DMA element is above 73%. Note that we used two PIN diodes to fully suppress the CELC radiation in the element-off state. We analyzed from numerical simulations that although one diode works well in a reverse biased state and the efficiency as well as gain can increase a little bit, however, it is the element-off state that is important to consider for DMA design to fully suppress the resonance effect of the CELC meta-element for effective 1-bit performance. When using a single forward-biased diode, the meta-element is not fully suppressed. With two diodes, the radiation is highly suppressed in a diode-on state, which is desirable. Furthermore, the two diodes maintain the element's symmetry which prevents the asymmetrical current flow and thus the distortion of the radiation pattern. As shown in Fig.13, the radiated power of the proposed DMA in the element-off state is below 0.05 W when both PIN diodes are on.

The peak realized gain (realized gain takes into account the input impedance mismatch loss as well as dielectric and conductor losses) of DMA in radiating state is 7.59 dBi at 60.68 GHz for open circuit CELC, and 7.07 dBi at 61 GHz with bias network and off-state PIN diodes. The gain of the DMA element in the non-radiating state is below -5 dBi, as shown in Fig. 14. Note that more than 11 dB difference in radiated power and gain is observed in element-on and -off states, which reveals the excellent performance of the proposed DMA element.

The radiation mechanism is further illustrated in Fig. 15 through field distribution. As shown in Fig. 15 (a and b),

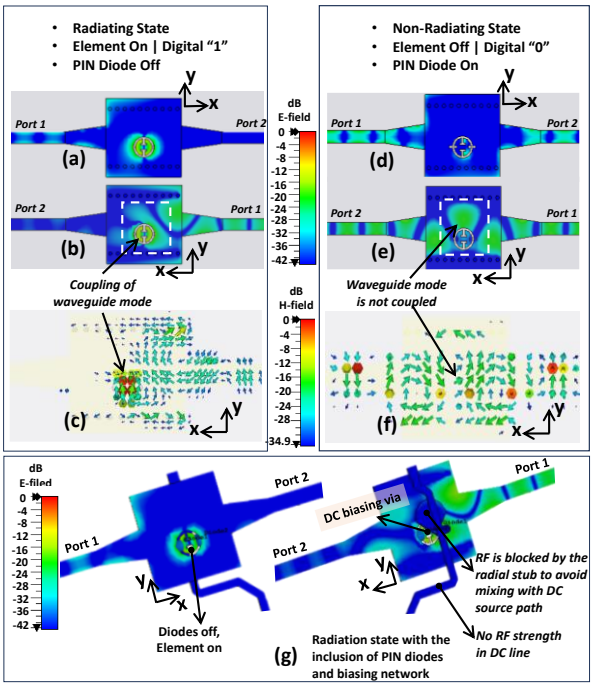


Fig. 15. E- and H-field distribution of DMA element at 60.5 GHz. (a) E-field top view in radiating state. (b) E-field bottom view in radiating state. (c) H-field bottom view in radiating state. (d) E-field top view in a non-radiating state. (e) E-field bottom view in a non-radiating state. (f) H-field bottom view in a non-radiating state. (g) Simulations of the realistic schematic model of the DMA element as shown in Fig. 6(b) reveal that radial stub effectively isolates RF from the DC source, and the field strength is the least towards the DC control line.

when PIN diodes are off, the CELC element couples with the waveguide mode at its resonant frequency of 60.5 GHz and radiates well with a strong E-field. Conversely, when PIN diodes are on, the CELC is shorted with the SIW structure and the waveguide mode does not couple with it, hence the energy travels towards port 2 with negligible radiated field intensity around it, as illustrated in Fig. 15(d, e). Similarly, H-field intensity in radiating and non-radiating states are shown in Fig. 15(c) and Fig. 15(f) respectively. The effect of radial stub and biasing via has minimal effect on the performance of DMA, as illustrated in Fig. 15(g).

C. POLARIZABILITY AND ARRAY FACTOR

Having a subwavelength dimension, the CELC resonator can be modelled as a polarizable magnetic dipole that couples with the waveguide mode to radiate EM energy. The induced dipole moment (m) in the presence of an in-plane magnetic field (H_i) is related to the polarizability (α) of the CELC resonator as [18], [42]:

$$m = H_i \alpha \tag{5}$$

Polarizability is a measure of how readily an external electric field may polarize a CELC resonator. Since the CELC resonator manifests typical Lorentzian-like resonance in its radiation state (as shown by |S21| dip of Fig. 11), therefore its polarizability can be modelled according to Lorentzian dispersion as:

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$$\alpha = \frac{F\omega^2}{\omega_0^2 - \omega^2 + j\omega\Gamma} \quad (6)$$

where ω is the excitation angular frequency, ω_0 is the resonant frequency which is associated with the fundamental geometrical features of each CELC resonator as $\omega_0 = \frac{1}{\sqrt{LC}}$, F is the coupling factor/oscillator strength, and Γ is the damping factor ($\Gamma = \frac{\omega_0}{2Q}$) due to resistive effect of the structure, where Q is the quality factor of the resonator ($Q = \frac{\sqrt{LC}}{R}$). According to (5) and (6), the field radiated from the CELC resonator has an amplitude and phase determined by the in-plane EM wave multiplied by the polarizability of the CELC resonator.

As the SIW is excited in dominant TE_{10} mode, therefore the transverse component of the magnetic field at the position of a given metamaterial element causes the predominant excitation which has sinusoidal variation as a function of the distance along the propagation direction. As the EM wave propagates along the x-axis in the designed case, the in-plane transverse magnetic field component along y-direction is expressed as:

$$\boldsymbol{Hi} = H_0 e^{-j\beta x_i} \boldsymbol{y} \quad (7)$$

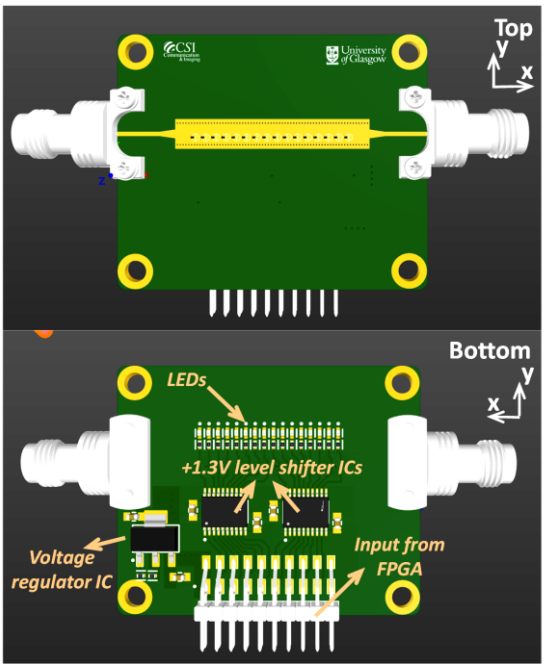
where H_0 is the initial magnetic field fed into the SIW structure, β is the effective waveguide propagation constant in SIW (which can be given as $\beta = \frac{\omega}{v_g} = \frac{n_g \omega}{c}$), and x_i is the position of the ith element in case of a DMA. The far-field pattern at a distance (r) of a CELC resonator is given by the far-field magnetic field vector $\boldsymbol{H_{rad}}$ as:

$$\boldsymbol{H_{rad}} = \frac{\omega^2 m}{4\pi r} cos\theta [e^{-jkr + j\omega t}] \hat{\boldsymbol{\theta}} \quad (8)$$

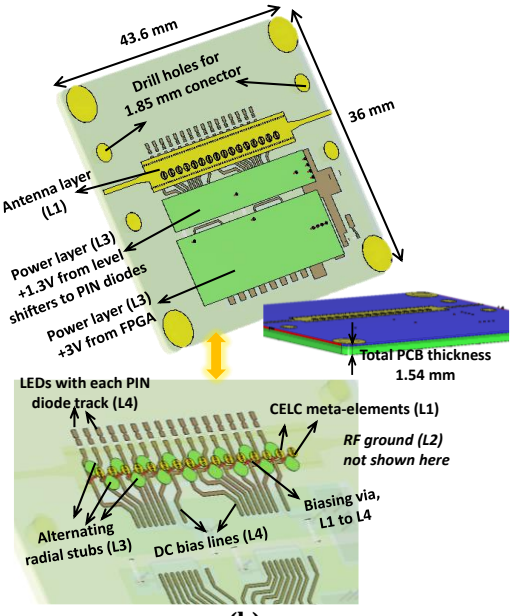
Where r is the approximated magnitude of the difference between the location of the radiation source and the observation point at a far-field distance, $\hat{\boldsymbol{\theta}}$ is the unit vector in the direction of the radiated wave indicating the direction of the magnetic field, and ($k = 2\pi/\lambda_0$) is the free-space wave number. Note that the radiation pattern of a single DMA element (as shown in the inset of Fig. 14) follows the cosine function as dictated by (8) where its intensity is towards broadside (i.e., towards 0°) with maximum intensity ($cos\ 0° = 1$) and diminishes around 90°. Considering that the elements do not strongly perturb the waveguide mode and do not interact with each other, the azimuth far-field radiation pattern of the 1-D DMA of N dipole elements in a desired direction (ϕ) from the superposition of the field radiated by each CELC radiator as:

$$\boldsymbol{H_{rad}} = H_0 \frac{\omega^2}{4\pi r} e^{-jkr} cos\theta \sum_1^N \alpha_i(\omega)\, e^{-jx_i(\beta + ksin\phi)} \hat{\boldsymbol{\theta}} \quad (9)$$

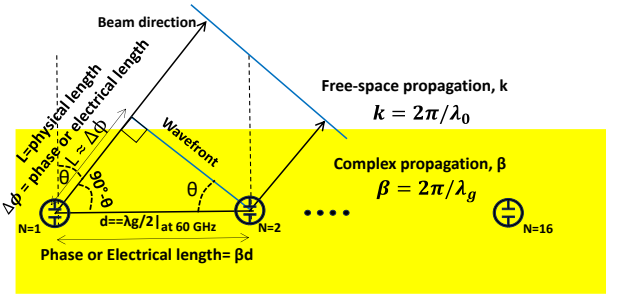
Where $cos\theta \sum_1^N \alpha_i(\omega)\, e^{-jx_i(\beta+ksin\phi)}$ is the array factor of 1-D DMA. Being 1-D array, the in-plane wave has the dependence on $e^{-jkxsin\phi}$ factor, ϕ is the angle of propagation normal to the surface of array aperture, and $\hat{\boldsymbol{\theta}}$ shows



(a)



(b)



(c)

Fig. 16. (a) Schematic view of the proposed 60 GHz 1-D DMA PCB. (b) Exploded schematic view of 4-layer PCB of DMA with biasing network revealed. (c) The schematic diagram for DMA beam-steering.

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directional wave vector. The terms in (9) account for the dipole strength, angular frequency, distance from the array to the observation point, the angular dependence of the radiation pattern, and the spatial distribution of magnetic dipole elements along the array. The array factor is a function of the geometry of the array and the excitation phase.

Note that fundamentally ω_0 relates to the inductance and capacitance of the resonant circuit in the usual manner and its resonance response can be controlled either through the geometry of the CELC element, modifying the local dielectric environment, or by integrating electronic components such as PIN diodes (as in this work) with each meta-element which will change its resonance and thus the phase. The phase accumulation in the guided wave introduces the phase variation required to produce directed beams by exciting CELC elements along the aperture. It is an interesting phenomenon that varying the separation distance and/or the phase β between the elements is achieved by the on and off scheme of PIN diodes and the characteristics of the total field and the beam shapes (beamwidth and number of lobes) of the array is dynamically controlled. In this way, dynamic reconfigurability of radiation patterns is achieved with extremely low power requirements and without the need for complex active circuitry.

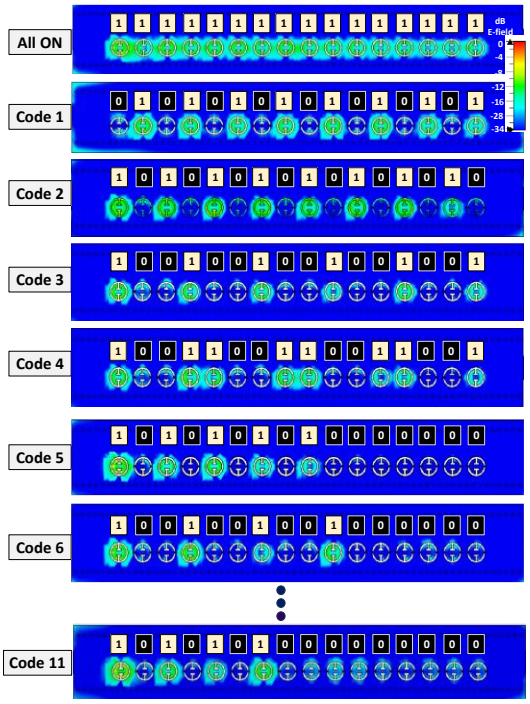
V. DMA ARRAY DESIGN AND SIMULATED RESULTS

A. PCB LAYER STACK UP OF THE PROPOSED DMA

The complete PCB view of the proposed DMA is shown in Fig. 16(a) and the perspective view of the interlayer schematic of the 4-layer PCB depicting the biasing scheme is shown in Fig. 16(b). The DMA comprises 16 CELC meta-elements, thus encompassing 32 PIN diodes. The dimensions of the SIW structure are the same as used to excite single DMA element, except that the length of SIW aperture is increased to 27.36 mm to accommodate 16 meta-elements (see Fig. 2). The S-parameters of elongated SIW structure are already shown in Fig. 3. The inter-element gap is 1.44 mm, which is equivalent to $\lambda_g/2$ or $\lambda_0/3.47$ at 60 GHz to avoid mutual coupling. Consequently, the radial stubs (at L3) connected to each biasing via are oriented in alternating directions to fit within the confined area. Moreover, L3 is exclusively designed to route control lines of PIN diodes. Initially, +3.3 V sourced from the FPGA output is directed to the input of two level-shifter ICs, each equipped with 8 output ports to control a total of 16 meta-elements. Subsequently, the output of the level shifter ICs is maintained at +1.3V (for diode-on case), serving as the supply to activate the PIN diodes. At L4, the path of each pair of PIN diodes is then serially connected to an LED, specifically designed for debugging and providing a clear visualization of the on/off states of the PIN diodes during the application of any digital coding sequence from the FPGA.

B. BANDWIDTH OF THE PROPOSED DMA

It is worth mentioning here that the characteristics of DMA such as input impedance, pattern, gain, side lobe level (SLL), etc., are critically influenced by the coding sequence.



Beamforming Coding Sequence	
Code 1:	0101010101010101
Code 2:	1010101010101010
Code 3:	1001001001001001
Code 4:	1001100110011001
Code 5:	1010101010000000
Code 6:	1001001001000000
Code 7:	0001000100010001
Code 8:	0000100001000010
Code 9:	0000010000010000
Code 10:	0111010101010101
Code 11:	1010101000000000

Fig. 17. Radiation state of the proposed DMA for different coding sequences at the respective frequency of maximum excitation of meta-elements. 11 different applied coding sequences are revealed here. 1's and 0's are w.r.t CELC element states, whereas inverting the same code will depict the state of PIN diodes.

Therefore, in addition to -10 dB input impedance bandwidth, the "pattern bandwidth" (i.e., gain, SLL, HPBW) should also be considered from a wireless communication point of view. A visualization of the simulated radiation state of the DMA for different coding sequences at the corresponding operating frequency of a maximum number of elements excitation is demonstrated in Fig. 17. Note that $2^{16} = 65536$ different combinations are possible from the coding space, however, to verify the prototype we present 11 selected coding sequences as demonstrated in Fig. 17.

While a single DMA element sets a distinct resonance frequency with a high Q-factor, the utilization of different coding combinations in case of DMA formation is expected to broaden the bandwidth by decreasing the quality factor. This phenomenon is primarily attributed to the relatively wider impedance matching level at the input port of the DMA across a specific frequency range due to different radiation states of meta-elements which arise because of dynamic coding sequences. Each unique code causes variations in the phase across the DMA aperture, resulting in the overall difference in impedance bandwidth (i.e.,

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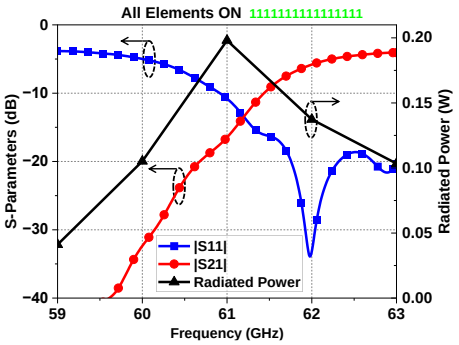


Fig. 18. S11, S21, and radiated power of DMA when all elements are in radiation state (i.e., all PIN diodes are reverse biased).

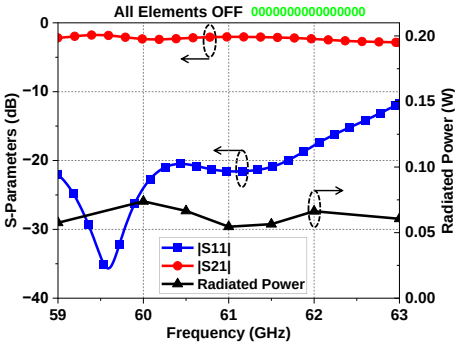


Fig. 19. S11, S21, and radiated power of DMA when all elements are in non-radiation state (i.e., all PIN diodes are forward-biased).

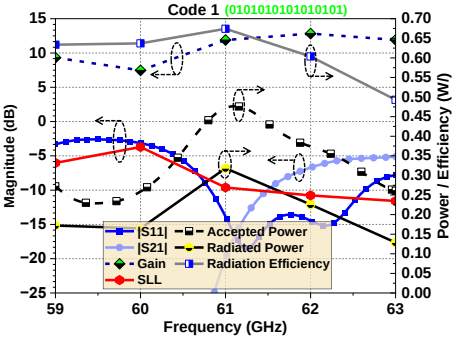


Fig. 20. S-parameters, radiated power, and gain of DMA for code 1.

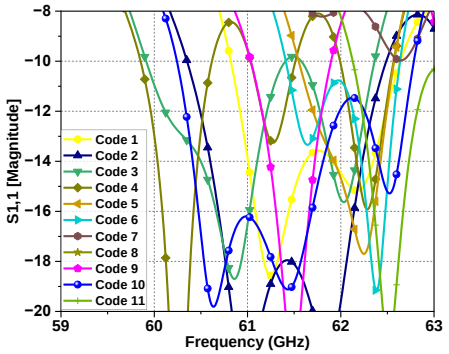


Fig. 21. Reflection coefficient of various coding sequences from 58 to 64 GHz.

reflection coefficient). As a result, a typically wider operational bandwidth is achieved in case of DMA array, as compared to that of a single antenna element.

The simulated |S11|, |S21| and radiated power of a fully radiating DMA (i.e., when all elements are in a radiation state) are shown in Fig. 18. This state of DMA has a maximum radiation power at 61 GHz, where |S11| < -10 dB and |S21| is around -16.4 dB. When all PIN diodes are on, the DMA is non-radiating. As expected, it manifests high |S21| and low |S11| with negligible radiated power, as shown in Fig. 19. However, this would be a trivial case where the radiation effect (antenna effect) of the DMA is lost.

To further elucidate the concept of operating bandwidth of the DMA, consider the input impedance and radiation profile for code 1, as shown in Fig. 20. The optimum operating region for code 1 spans from 60.7 GHz to 63 GHz, where |S11| < -8.5 dB, SLL is below -11 dB, realized gain is above 10 dBi with peak gain of 12.82 dBi at 62 GHz, and peak radiation efficiency is 67.42% at 61 GHz. Note that around 58.5 GHz, although |S11| is < -10 dB and gain is about 10 dBi, however SLL increases to -7 dB. At 60 GHz, the reflection coefficient, SLL, and total efficiency degrade, however, the radiation efficiency is above 63%. Similarly, between 63 GHz and 64 GHz, |S11| < -10 dB, gain is above 10.5 dBi and SLL is at -11 dB, however, the radiation efficiency drops below 50% in this region (because radiated power is low). Therefore, these performance tradeoffs decide the choice of operating bandwidth depending on required application scenarios. For mmWave holographic imaging applications, code 1 might perform quite well from 59 GHz to 63 GHz, however requirements of wireless communication domain are usually stringent regarding SLL, impedance matching, and radiation efficiency, therefore optimum bandwidth might be considered differently.

The reflection coefficient of various selected codes with maximum selected |S11| < -8 dB level is shown in Fig. 21. Although, some coding sequences show quite good gain, SLL, and efficiency performance at 59 GHz and lower the performance as the operating frequency increases beyond 62 GHz, nevertheless, the high-performance operating range for the proposed DMA lies between 60 GHz and 63 GHz, showcasing a bandwidth exceeding 2.5 GHz. This surpasses the 2.16 GHz bandwidth of a single channel at the 60 GHz ISM band under IEEE 802.11ad and IEEE 802.11ay protocols. Onwards in this paper, we present simulated and measured results at 60 GHz, 61 GHz, and 62 GHz to demonstrate the performance of the DMA prototype.

C. PROGRAMMABLE ELECTRONIC BEAM-STEERING WITH DIGITAL CODING

Owing to linear array topology, the proposed DMA offers fan-shaped beams with relatively narrow beams in the x-z plane. The 3-D radiation patterns for various codes along with heatmaps are shown in Fig. 22. The principal plane for beam-steering (and thus the plane of interest) is x-z plane, along the arrangement of CELC meta-elements, as elucidated in Fig. 23. The radiation patterns are wider and mostly directed towards 0° in y-z plane. For the sake of completion, 2-D radiation patterns at 60, 61 and 62 GHz for some codes in the y-z plane are shown in Fig. 24.

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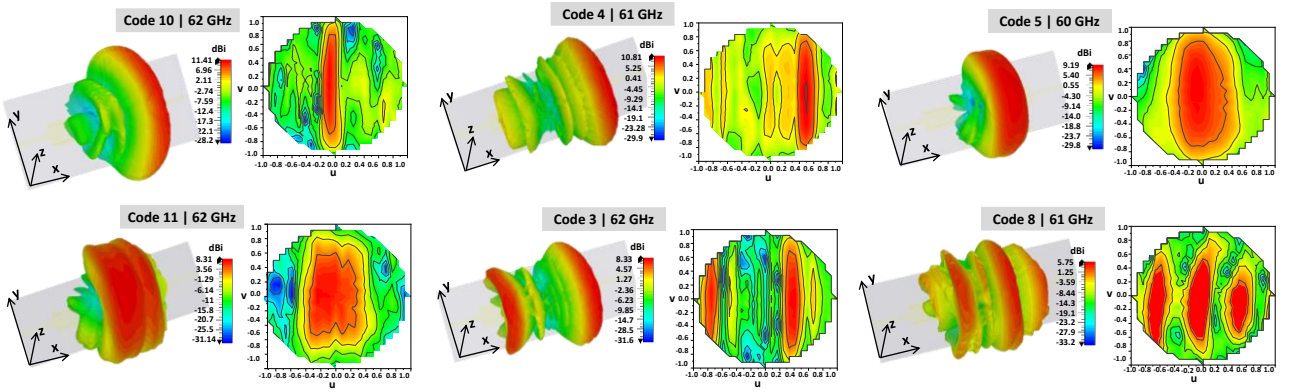
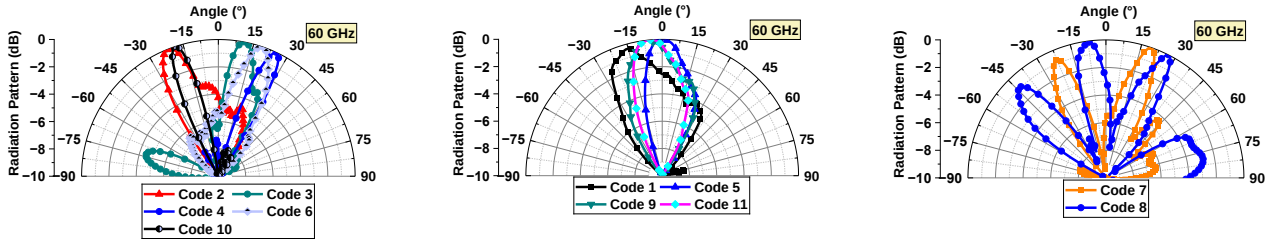


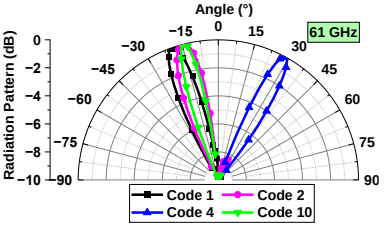
Fig. 22. Illustration of various types of simulated 3-D radiation patterns along with corresponding u-v orthographic heat maps produced from different digital coding combinations.



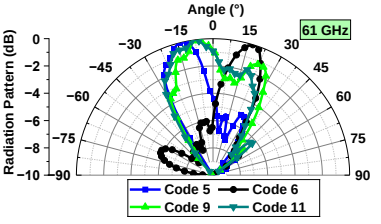
(a) Beams with narrow HPBW at 60 GHz

(b) Beams with wider HPBW at 60 GHz

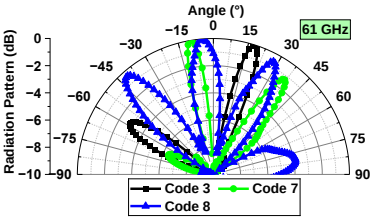
(c) Multiple directed beams at 60 GHz



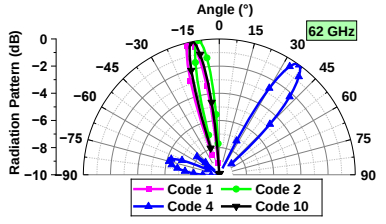
(d) Beams with narrow HPBW at 61 GHz



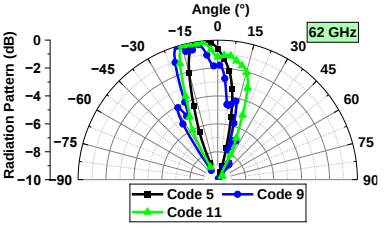
(e) Beams with wider HPBW at 61 GHz



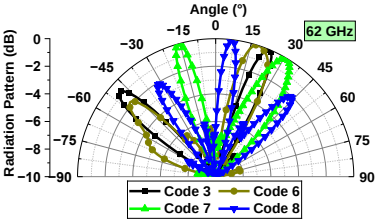
(f) Multiple directed beams at 61 GHz



(g) Beams with narrow HPBW at 62 GHz



(h) Beams with wider HPBW at 62 GHz



(i) Multiple directed beams at 62 GHz

Fig. 23. Simulated beam-steering plots of DMA at 60 GHz, 61 GHz, and 62 GHz using different coding sequences in the x-z plane. (a, d and g) show directed beams with narrow HPBW. (b, e, h) show beams with relatively wider HPBW.(c, f, i) show multiple beams.

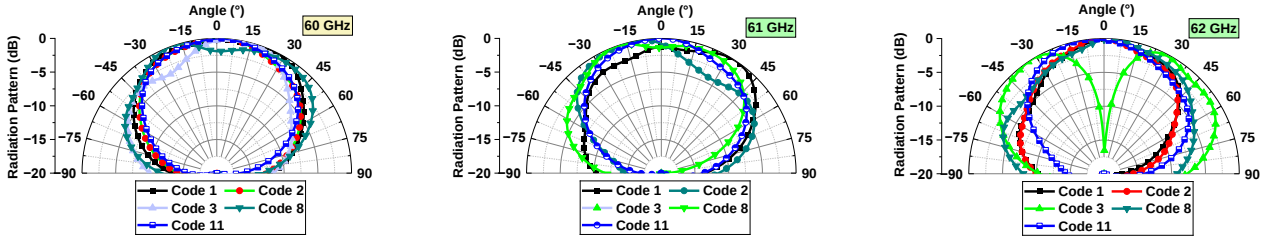


Fig. 24. Simulated radiation pattern of the proposed DMA in y-z plane at 60 GHz, 61 GHz, and 62 GHz for different coding combinations.

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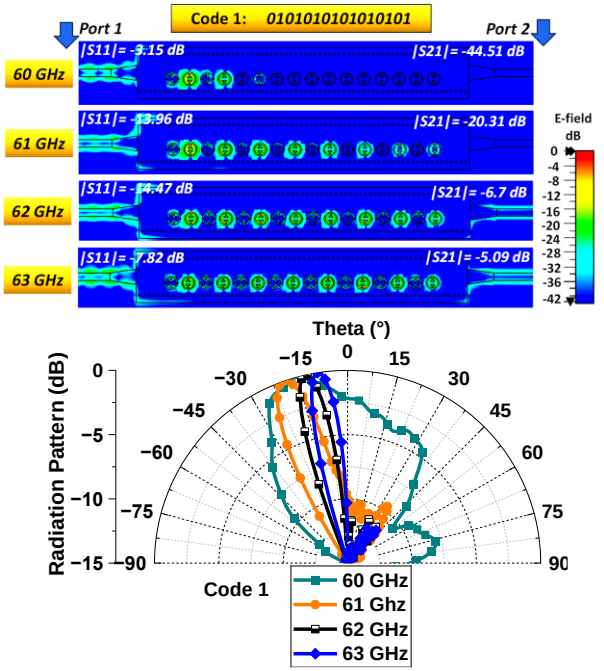


Fig. 25. Response of the DMA for code 1 leading to radiation pattern with varying HPBW.

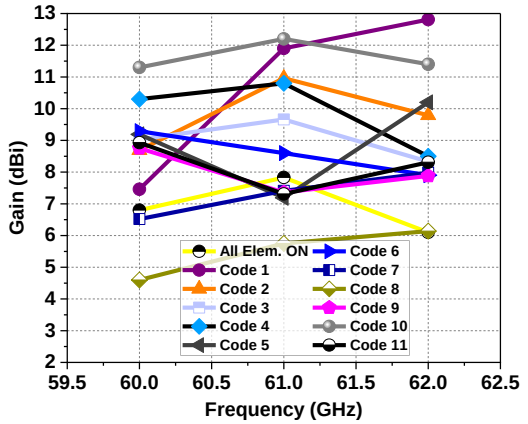


Fig. 26. Simulated realized gain of the DMA for various beamforming codes.

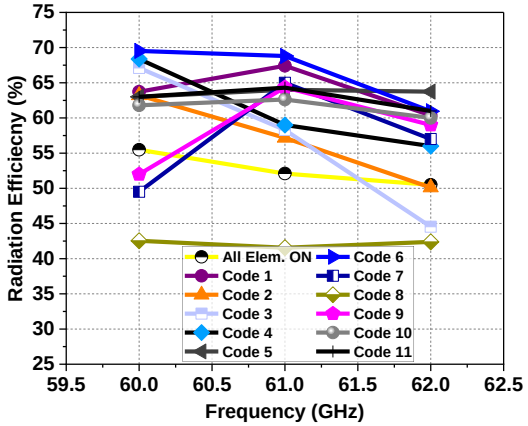


Fig. 27. Simulated radiation efficiency of the DMA for various beamforming codes.

Different coding combinations produce different effective radiation apertures of the DMA and form dynamic beam shapes, as illustrated in Fig. 23. We can categorize the radiation patterns into three different types for different coding sequences at each operating frequency, such as beams with narrow half-power beamwidth (HPBW), beams with relatively wider HPBW, and multiple beams. Code 2, code 4, and code 10 produce narrow directed beams with an average HPBW of about 10° at 60 GHz, 61 GHz, and 62 GHz. Code 5, code 9, and code 11 produce beams with relatively wider HPBW of around 33° at these three frequency points. Code 7 and code 8 produce multiple beams at 60 GHz, 61 GHz, and 62 GHz.

The observed beamsteering range is ±50° (Fig. 23(f and i)) for the tested coding space (may increase for other codes). The radiation pattern with code 1 produces wider HPBW of 38° at 60 GHz, whereas at 61 and 62 GHz, it shows relatively narrow HPBW of 12.2° and 8.9° with SLL of -10.35 dB and -11.20 dB respectively. Code 5 provides a broadside beam at 60 GHz whereas all-elements-on case provides a broadside beam at 62 GHz. The beam-steering range for the 11 tested codes varies within ±45°. The beam diversity of DMA is further explained in the next sub-sections.

The gain of the DMA for different coding combinations within 60 to 62 GHz is provided in Fig. 26. For code 1, code 2, code 4, and code 10, the gain is above 10 dBi at 61 GHz with SLL less than -10 dB. For multi-beam codes such as code 8 and code 9, each lobe has a minimum gain of 5 dBi. Moreover, the radiation efficiency is above 60% for most of the coding sequences as shown in Fig. 27. The DMA is linearly polarized and shows high cross polar rejection.

D. BEAMWIDTH SYNTHESIS AND VERSATILITY OF DMA

The proposed CELC meta-element exhibits interesting behavior when stimulated in a multitude of array topologies in a waveguide aperture. Because the polarizability of the CELC meta-elements is greatly contingent on the resonance frequency as governed by (6), their superposition manifests a dynamic behavior at different frequencies corresponding to different digital coding sequences. Consequently, this leads to the formation of versatile beam morphologies. To elucidate this, consider Fig. 25, where the radiation states and beamforming are demonstrated for code 1 at 60 GHz, 61 GHz, and 62 GHz. Since a resonating CELC element behaves as an open circuit at its resonance frequency, hence for code 1, CELC 2 and 4 appear to be strongly resonating while leading to reduced wave transmission towards the output port (|S21 ≈ -41.5 dB) and presenting high impedance mismatch towards the input port (|S11| ≈ 3.15 dB). In this way, other expected on-state elements do not find enough coupling and therefore the effective radiating aperture ($L_{effective}$) of the DMA is reduced. The beamwidth of a uniform linear array antenna is inversely related to the effective radiation aperture of the antenna array and can be estimated as [43]:

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$$HPBW° \approx \frac{0.886\lambda}{L_{effective}} \quad (10)$$

From (10), it can be intuitively deduced that as the effective length of DMA aperture at 60 GHz is reduced, the HPBW is substantially broadened. From 61 GHz to 63 GHz, the CELC elements couple relatively weakly being a function of resonant frequency from (6), and in turn more energy travels towards the outport port. In this way, a greater number of on-state CELC meta-elements tend to radiate as illustrated in Fig. 25. Therefore, the effective radiating aperture is increased at these operating frequencies, leading to beams with narrow HPBW. Narrow beams are applicable for fine grain resolution whereas wider beams are applicable for wide area coverage.

E. MULTI-BEAM GENERATION FROM DMA

The complex electromagnetic behavior of the DMA manifests itself in a variety of ways for different coding sequences and offers remarkable features through wave manipulation by binary coding. From Fig. 16(c):

$$cos\ (90° - \theta) = sin(\theta) = \frac{\Delta\phi}{\beta d} \quad (11)$$

$$\theta = \sin^{-1}\frac{\Delta\phi}{\beta d} \quad (12)$$

where $\Delta\phi$ is the phase difference (electrical length) between two consecutive CELC elements and θ shows the beam direction. The domain of $\sin\theta$ is the set of real numbers and its range varies between +1 and -1. Therefore, the inverse sine function ($\sin^{-1}$) only produces real solutions for the arguments bound within +1 and -1. Outside this domain of inverse sine function, the solution is not real. As the phase in a waveguide medium is periodic and repeats after every 2π rotations, thus in general, we can generalize $\Delta\phi$ with $\Delta\phi + 2n\pi$, where n=0, ±1, ±2, ±3... Thus,

$$\theta = \sin^{-1}\left[\frac{\Delta\phi+2n\times\pi}{\beta d}\right] = \sin^{-1}\left[\frac{\Delta\phi+2n\times\pi}{2\pi}\times\frac{\lambda_g}{d}\right] \quad (13)$$

From (13), the grating lobe suppression theory of phased array antenna governs that if the modulus of $\frac{\Delta\phi+2n\times\pi}{2\pi}\times\frac{\lambda_g}{d}$ is greater than 1 for all n≥1, the solution will be non-real values which can be ignored, and the grating lobes can be avoided. This will happen if $\frac{\lambda_g}{d} > 1$ which implies $d < \lambda_g$, or most commonly d < $\frac{\lambda_g}{2}$ to avoid grating lobes. However, if $\frac{\lambda_g}{d} < 1$, (the case when d > λ_g), then (13) gives multiple real solutions for some $n > 0$ which translates into grating lobes (multi-beam scenario). Moreover, at higher frequency points such as 62 GHz, the electrical inter-element distance (λ_g) seems greater than that of 60 GHz, therefore multi-beams tend to be more pronounced at higher frequency end, such as for code 3.

For conventional phased array antennas, the physical distance between the antenna elements (d) cannot be changed once the array is designed (as every element is always in a radiation state and contributes to effective antenna aperture). Contrary to this, the hallmark effect of the DMA is that we can dynamically control the radiation state of individual meta-elements and thus can manipulate the inter-element distance in real-time using a particular digital coding sequence. Therefore, exploiting this effect, we applied code 7 and code 8 to purposely increase the inter-CELC distance (d). Note that the notion of multiple beams from a metasurface antenna's perspective is usually different from the grating lobe perception of conventional phased array antennas. Therefore, thigh gain multiple beams (from grating lobes) are predicted as a blessing, rather than a curse. The multi-beam scenario of the proposed mmWave DMA can be employed in indoor IIoT applications or multi-user scenarios such as in smart factories where the network is usually highly secured and multiple users can be connected simultaneously through a single antenna.

VI. PROTOTYPE FABRICATION AND MEASUREMENT RESULTS

The fabricated prototype of the 16-element DMA along with the integrated FPGA is shown in Fig. 28 (a). The size of the fabricated DMA array is 43.6 mm × 36 mm × 1.54 mm. Since the DC biasing path of each CELC meta-element is equipped with an LED at L4, therefore the correct response of the digital coding sequence generated through FPGA is readily noticeable through the on and off states of LEDs (corresponding to on and off states of PIN diodes), as demonstrated in Fig. 28 (c). A graphical user interface (GUI) is exclusively designed to generate any of the desired binary coded combinations and a software program is pre-loaded in the FPGA to reconfigure the DMA, as shown in Fig. 28 (d). The bias voltage applied to each PIN diode can be changed by controlling the FPGA, such that the ON/OFF state of the PIN diode together with the radiation state of each CELC element can be controlled.

For ease of measurement and rapid prototype verification of the designed CELC meta-element, a separate prototype of the single DMA element was also fabricated without a biasing network as shown in Fig. 28 (b). This depicts a radiating DMA element state (being an open circuit CELC element). The off-state (non-radiation) performance was instantly accomplished by merely shorting the capacitive gap between CELC and SIW aperture (analogous to PIN diode switching on) through soldering.

A. S-PARAMETER MEASUREMENTS

The S-parameters of the single DMA element as well as the DMA array for various beamforming codes were measured using Agilent E8361A vector network analyzer (VNA) after 2-port calibration, as demonstrated in Fig. 28 (e). The resonance frequency of the single DMA element is around 61.6 GHz as shown in Fig. 29. Note that the resonance trend (curve behavior) is quite similar to the simulated results, however, the measured resonance frequency is blue-shifted due to practical fabrication tolerances. The measured group delay is negative around 61.6 GHz as shown in Fig. 30.

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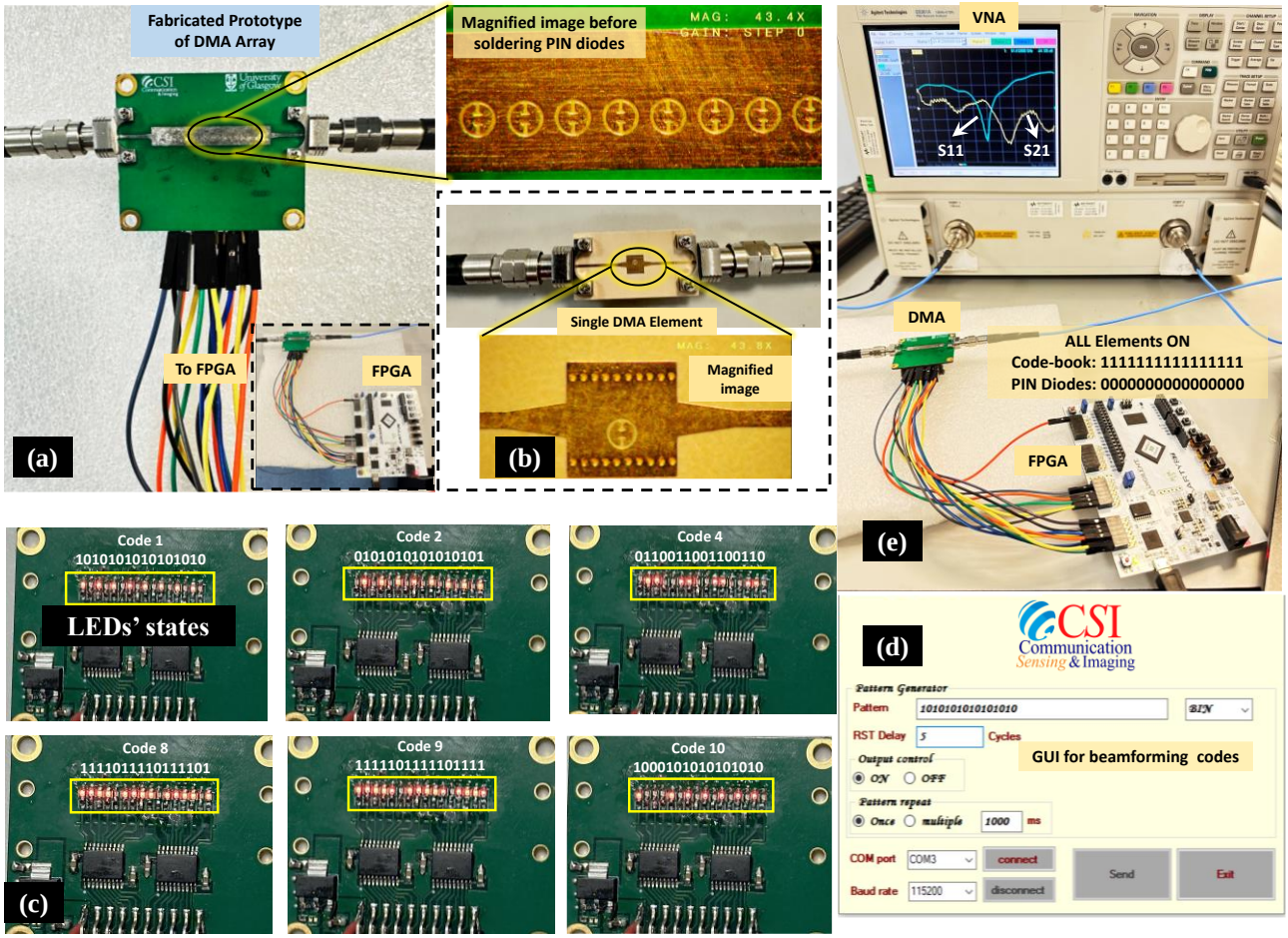


Fig. 28. (a) Fabricated prototype of DMA integrated with FPGA along with a magnified image of the top layer. (b) fabricated single DMA element prototype and its magnified image. (c) Practical visualization of different on/off states of LEDs corresponding to different coding combinations. The coding 1's and 0's shown in this image correspond to the on and off states of PIN diodes respectively (i.e., inverted to the codebook sequence of Fig. 17). (d) Developed GUI to generate any of the desired beamforming coding sequences for FPGA towards DMA. (e) Measurement setup for S-parameters on VNA.

The measured transmission phase of the single DMA element shows a positive slope around the resonance frequency and therefore is a confirmation of the negative group delay. Hence, it reveals left-handed metamaterial properties as described in Section III above.

Before measuring the reflection and transmission coefficients of the DMA array, the DC test was conducted to ensure the correct operation of the DC biasing network with FPGA. The measured results S-parameters are stable during prototype movement and match quite well with the simulations, thereby confirming the robust response of the fabricated prototype. The measured S11 and S21 during fully radiating state (when all elements are in radiation state) are shown in Fig.31. Moreover, for various coding combinations, the reflection coefficients were measured and are shown in Fig. 32.

Minimal deviations in measured S-parameters are mainly due to the practical response of diodes at the 60 GHz band, soldering effects, fabrication tolerance of PCB, and practical losses in the conductor and dielectric. Nevertheless, the resonance behavior of the measured results is in great agreement with the simulation results in general.

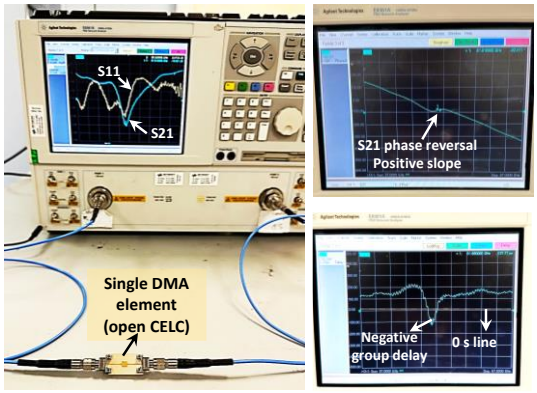
B. RADIATION PATTERN AND GAIN MEASUREMENTS

The measurement setup for the radiation pattern is shown in Fig. 33. A standard gain V-band horn antenna was used as a transmitter and was mounted on a turntable capable of rotating 360°. The turntable is equipped with a programmable motor which is automated and controlled through LabView. The horn antenna was rotated over a span of 180° (-90° to +90°) with 1° angular steps. The DMA was fixed at a far-field distance of 40 cm ($>2D^2/\lambda$, where D is the largest dimension of DMA) from the horn antenna. A single-tone continuous-wave (CW) frequency signal was produced at each desired operating frequency of 60 GHz and 62 GHz to record the radiation pattern through VNA over a span of 180°.

The radiation patterns at 60 GHz and 62GHz for different beamforming codes in the azimuth plane (x-z plane) were measured, as shown in Fig. 34 and Fig. 35 respectively. The measured patterns show good agreement with the simulated results, while the measured SLL and null depths are within 1.5 dB as compared to the simulated ones. Due to linear geometry, the radiation patterns in the elevation plane possess wide beamwidth and are symmetrical mostly

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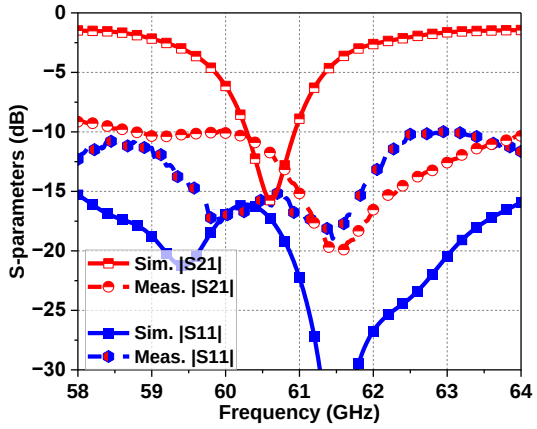


Fig. 29. (Top) S-parameter measurements of single DMA element on VNA. (bottom) Measured reflection and transmission coefficient of single DMA element in radiating state.

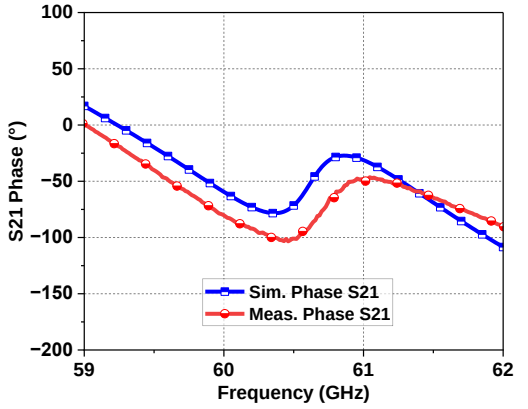


Fig. 30. Measured transmission phase of the single DMA element in radiating state confirming negative group delay.

towards broadside, thus are not of interest for measurements (presented in Fig. 24 above).

The gain was measured with the relative gain comparison method using a linearly polarized standard gain horn antenna with known gain ($Gain_{horn}$). Two sets of measurements

were performed: one with horn antenna and the other replacing it with the DMA at the same frequency and identical setup. The peak S21 levels were recorded for both sets and then the realized gain of the DMA was calculated by solving two sets of simultaneous Friis equations and using the final equation as [43]:

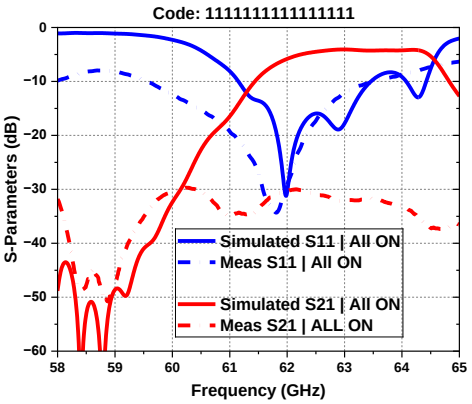


Fig. 31. Measured reflection and transmission coefficients of the DMA array when all elements are in radiation state (i.e., all PIN diodes are OFF).

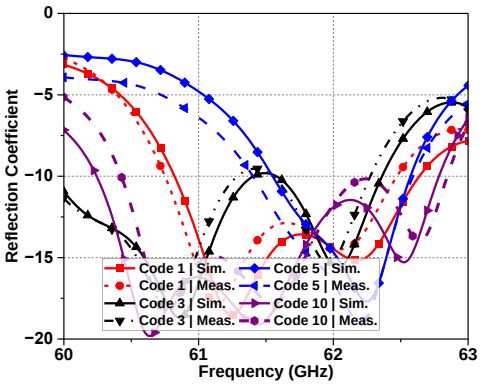


Fig. 32. Measured reflection coefficient of DMA array for various coding sequences.

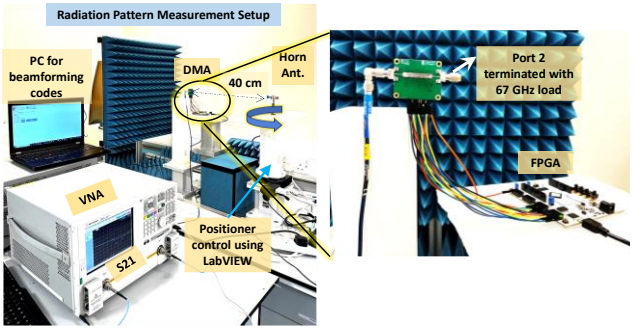


Fig. 33. Over-the-air experimental setup to measure radiation pattern and gain.

$$Gain_{DMA} = Gain_{horn} + S21_{DMA} - S21_{horn} \quad (14)$$

Here, $S21_{DMA}$ and $S21_{horn}$ are the S21 levels (in dB) directly measured from the VNA at the desired frequency points and at maximum power level. The trend of gain variation for different beamforming sequences across the frequency band of interest matches well with the simulated gain result, as shown in Fig. 36. The peak measured gain is 11.63 dBi at 62 GHz for code 1. The difference between simulated and measured gain is mainly due to practical conductor losses, PIN diodes, and RF connector effects.

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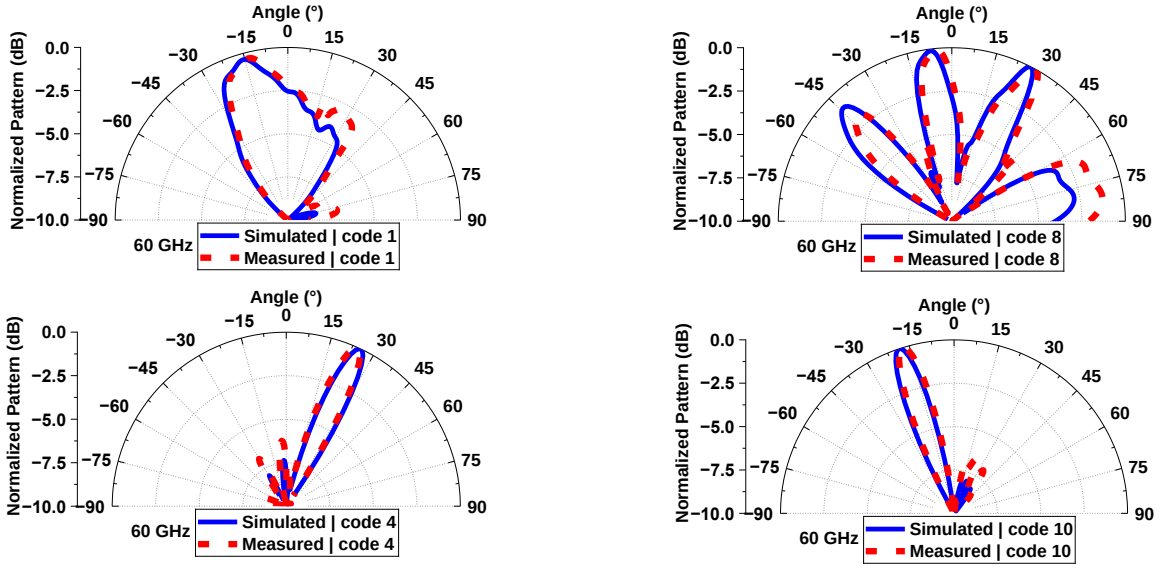


Fig. 34. Measured radiation patterns for different beamforming coding combinations at 60 GHz in the azimuth (x-z) plane.

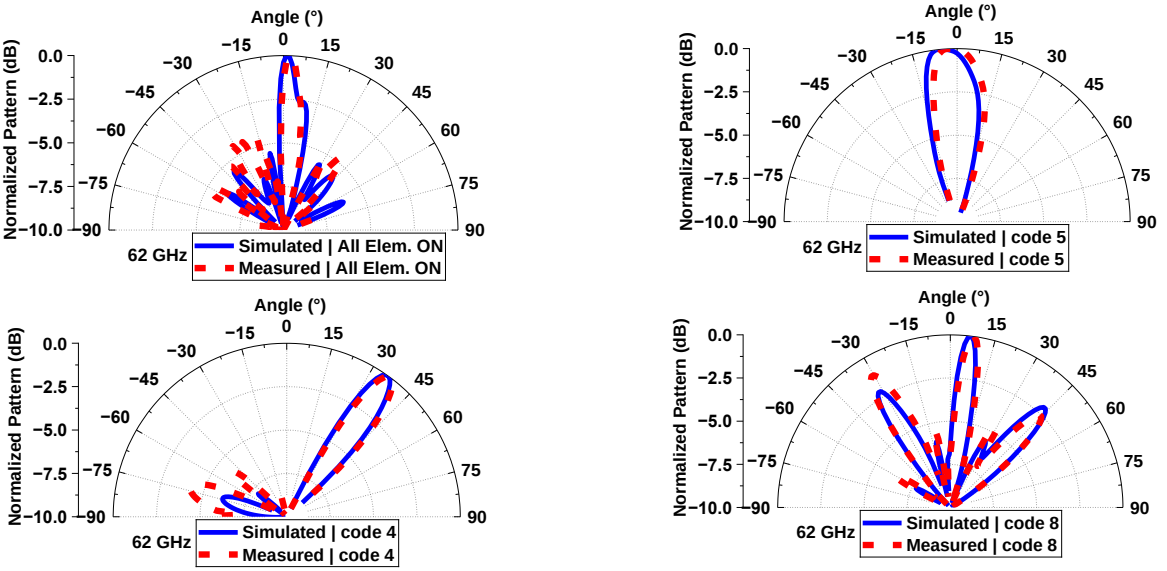


Fig. 35. Measured radiation patterns for different beamforming coding combinations at 62 GHz in the azimuth (x-z) plane.

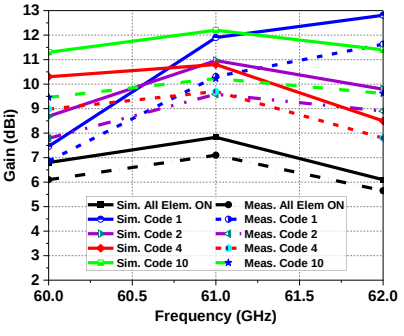


Fig. 36. Measured gain across frequency for different beamforming coding combinations.

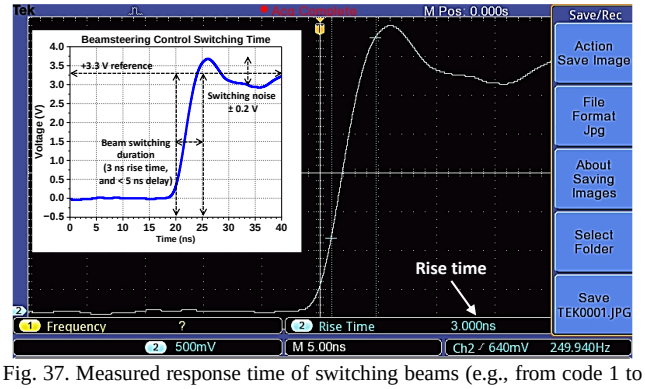


Fig. 37. Measured response time of switching beams (e.g., from code 1 to code 2) from FPGA.

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Table. I. Comparison of state-of-the-art DMAs in the literature.

Ref.	Operating Frequency Region (GHz)	DMA Geometry	Electronic Switching Component	DMA Aperture Size (L: mm × W: mm)	Observed Peak Gain (dBi)	Observed Peak Radiation Efficiency (%)
[14], [15]	11.8	4 × 16 planar	PIN Diode	152 × 60.5	10	Not available
[31]	12	1 × 16 linear	PIN Diode	124 × 15	11.02	68.36
[32]	10	8 × 12 planar	Varactor Diode	229 × 120	10	11
[33]	23.5	1 × 41 linear	PIN Diode	Not available	9.3	41.7
[34]	26	8 × 16 planar	Varactor Diode	100 × 89.4	7	Not available
[59]	21	Planar random aperture	PIN Diode	114.24 × 114.24	7.72	40.55
This Work	**60**	**1 × 16 linear**	**PIN Diode**	**27.36 × 4.2**	**11.63**	**69.52**

C. BEAMFORMING RESPONSE TIME AND AGILITY MEASUREMENT

The DC output of FPGA is +3.3 V which is converted to +1.33 V using an on-board voltage regulator IC to switch on the PIN diodes (element off state), whereas 0V is used to switch off the PIN diodes (element on state). The control signal from FPGA is operated at 250 MHz clock frequency with parallel LVCMOS 3.3 IO standard to achieve a high beam-switching update rate.

All PIN diodes are parallelized thereby the digital coded signal is simultaneously provided to all elements which significantly reduces the beamforming switching time and offers low-latency beam-steering. Therefore, the overall agility (latency of the antenna system) of the beam-switching between any two coded combinations (for pattern reconfigurability) is around 5 ns. The DMA electronically switches the beam from one direction to a specific scan angle through FPGA, and a mixed signal oscilloscope (TBS1072B) was used to observe the signal level under switching conditions from code 1 to code 2. The time between two steady states is considered as the beam-switching time, which was measured to be less than 5 ns as experimentally demonstrated in Fig. 37.

A comparison summary of the closely related DMA in the literature is presented in Table I. The proposed DMA design offers more than 2.16 GHz of -10 dB impedance bandwidth around 60 GHz, above 8 dBi gain for various beamforming codes and high radiation efficiency of above 60%.

VII. POTENTIAL USE-CASES OF THE PROPOSED 60 GHZ DMA

The real-time agile beam tailoring and signal processing capabilities of DMAs with simplified transceiver hardware makes them a potential enabler for next-generation mmWave communication, sensing, and imaging (CSI) applications. Furthermore, since the 60 GHz band is an unlicensed spectral region, it offers applicability to a wide range of industrial and healthcare applications. In this section, we briefly highlight the potential use cases of the proposed mmWave DMA in CSI avenues, offering new opportunities and open research challenges towards innovative applications due to its flexibility, dynamic reconfigurability, and license-free mmWave operating band.

A. COMMUNICATION

DMAs offer an innovative programmable wireless communication paradigm through software controlled antenna aperture [44]. Owing to the small size of meta-elements involved in the mmWave DMA, multitude of meta-elements in linear (1-D) topology form a subarray which behaves as a single antenna, and its corresponding planar topology can thus be employed for efficient hybrid beamforming. The 2-D planar arrays are envisioned to be utilized in massive-MIMO (multiple-input-multiple-output) base station and access point communication network [11], [21]. With relatively simple signal processing algorithms, a massive-MIMO system with a multitude of meta-elements can yield significant diversity gain and spectral efficiency, which is one of the key technologies for 6G and beyond [45].

The power consumption of a DMA-based system can be substantially lower as DMAs do not require as many RF chains or traditional active phase shifters to operate. Moreover, electronic beamsteering and multi-beam synthesis capability of the proposed DMA envisions to

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support large-scale 60 GHz indoor IoT networks encompassing high transmission rates and massive data throughput. An antenna system equipped with beam scanning capabilities can mitigate interference by selectively targeting the beam in a specific direction, enhancing the stability of data transmission and simultaneously optimizing spatial power allocation based on the node's communication data volume [26]. This optimization presents an opportunity to decrease the data collision rate, average energy consumption, and packet transmission delays. Nodes equipped with beamsteering antennas exhibit an 88% reduction in energy usage and a 24% decrease in data collision when compared to solutions using omnidirectional antennas [46].

Another potential application of DMAs is looming towards next-generation near-field beam focusing and near-field communication (NFC), which is a prospective research area [47]. Moreover, besides wireless communication, the wireless power transfer (WPT) through energy beamforming for IoT using DMAs has also been presented to be an efficient solution in the form of hybrid beamforming, as compared to fully digital beamforming networks [48].

The proposed 60 GHz mmWave DMA prototype with reduced substrate losses, high gain, well controlled SLLs, and above 65% radiation efficiency can be a potential candidate to explore and verify its applicability for the enhanced energy beamforming optimization scenarios as compared to expensive high-frequency fully digital beamforming solutions. This is certainly a potential research area that requires further experimental verification.

B. SENSING

DMAs are greatly envisioned to boost RF sensing accuracy in sensing and localization applications by employing antenna pattern diversity [19]. The object wave and reference propagating wave upon a DMA render it a reconfigurable holographic surface to synthesize a wide range of antenna patterns through externa programming. This adaptability is a key enabler to select specific patterns matched to the desired sensing scenarios [24], [49]. Machine learning based intelligent algorithms are possible to be devised to work with the proposed DMA prototype to identify the required set of antenna patterns for specific activities and then configure the DMA to synthesize those patterns.

Furthermore, the 60 GHz band is a potential mmWave Wi-Fi band with various working protocols such as IEEE 802.11ad/IEEE 802.11ay [50]. The proposed 60 GHz DMA prototype is fully compatible to work with these protocols to achieve mm-level fine-grained resolution for accurate sensing. The real-time electronic beamsteering property of the proposed DMA can be utilized for beamforming training using various algorithms to assess the Channel Impulse Response (CIR) for each direction of beamforming. to enable indoor mmWave Wi-Fi-based radar sensing. Recent studies have indicated the feasibility of employing sector-level sweeping procedures for opportunistic 60 GHz radar sensing in indoor environments, with a focus on smart IoT applications [51].

C. Imaging

The dynamically coded metasurface apertures have been widely utilized for microwave imaging applications [23], [25], [52]–[57]. Usually, imaging techniques involve frequency-swept signals where the large spectral bandwidth is typically required in synthetic aperture radar systems to resolve objects in range, which often requires costly and complex RF components. However, the DMA offers high spatial diversity at single/narrow frequency band with simplified programmable hardware.

A DMA can illuminate a scene with dynamic radiation patterns (where each DMA configuration is called as a “mask”) that can be altered based on the dynamic coding sequences. It is instructive to mention here that for imaging applications, random placement of meta-elements can also be utilized in DMA design, because SLL is not of the main concern (as opposed to communication regime where Nyquist limit of $\lambda_0/2$ is desirable for element placement). Thus, a sequence of pseudorandom radiation patterns interrogates a scene and the measurements of the return signal against each coding sequence can be acquired. Eventually, the scene can be reconstructed using computational imaging methods [58], [59].

For more fine-grained and high-resolution mmWave holographic computational imaging, the proposed 60 GHz DMA prototype is envisaged to hold a significant potential due to its versatility and control over beam-shaping features.

VIII. CONCLUSION

This paper proposes a programmable DMA with agile electronic beam-steering at 60 GHz mmWave ISM band. A new CELC metamaterial element is first designed and fed through the in-plane magnetic field of a low-loss SIW aperture. The left-handed metamaterial properties of the proposed meta-element are highlighted. The DMA element is integrated with a couple of PIN diodes whose biasing state can readily be changed with digital coding which manifests a fully addressable digital dynamic element. Then a 16-element DMA is designed using 4-layer PCB technology and fully controlled through a high-speed programmable FPGA for electronic beam-steering. The proposed array can synthesize different types of beam shapes through various digital coding sequences. The simulations and measurement results show good agreement. The proposed DMA design offers the benefits of low-cost, low-power, and ease of RF feed through edge fed technique and is a potential enabler to unleash futuristic CSI applications.

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This article has been accepted for publication in IEEE Open Journal of Antennas and Propagation. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/OJAP.2024.3386452

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This article has been accepted for publication in IEEE Open Journal of Antennas and Propagation. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/OJAP.2024.3386452

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This article has been accepted for publication in IEEE Open Journal of Antennas and Propagation. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/OJAP.2024.3386452



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This article has been accepted for publication in IEEE Open Journal of Antennas and Propagation. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/OJAP.2024.3386452



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